SOC 527

SOCIAL NETWORKS

Exponential Random Graph Models

Inferential Network Models Guest Lecture Series

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Lecture Outline

- Exponential random graph models (ERGMs)
- Empirical application
- Assessing model goodness-of-fit and degeneracy



Source: https://freshspectrum.com/evaluation-cartoons/

Lecture Goals

1. Have a general understanding of the exponential random graph model (ERGM).

2. Be able to interpret the results of a fitted ERGM.

3. Be able to diagnose the goodness-of-fit (GOF) and access the degeneracy of a fitted ERGM.



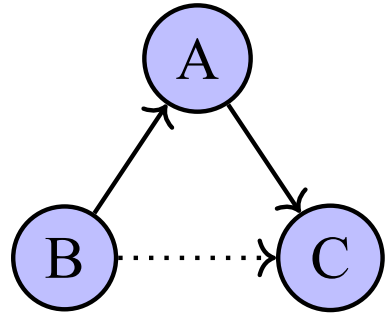
Why use exponential random graph models (ERGMs)?

Following Robins et al. [2007], we can use ERGMs to:

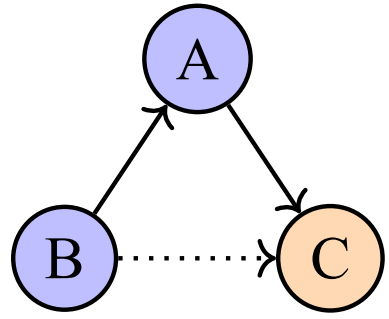
- → test if certain network configurations (substructures) occur more often in the observed network than expected by chance.

- → test competing hypotheses about how different social processes lead to the same network configuration.
  - ☞ Network clustering may be a outcome of "actors matching on exogenous attributes and actors forming partnerships on the basis of existing shared partners." [Hunter et al., 2008a, 252]

Transitive triads:
***f*(homophily)**



***f*(transitivity)**

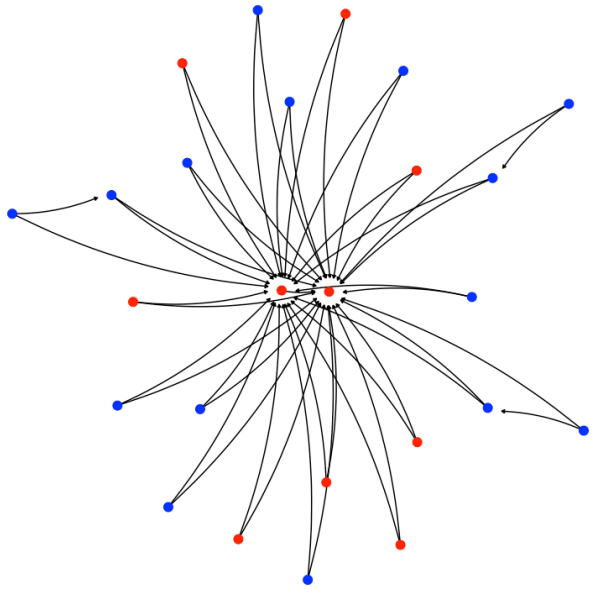


Why use exponential random graph models (ERGMs)?

Following Robins et al. [2007], we can use ERGMs to:

- → account for the dependence across dyadic ties which other statistical models cannot.
- → simulate networks under different research and counterfactual designs.
- → understand how (and if) social processes and local structures lead to the emergence of global network patterns [Duxbury, 2024]
  - ☞ Ex.: transitivity, homophily, sociality, popularity.

Scale-free network



Preliminaries on social networks

We assume that for each i and j actor in a set of N actors, there is a random variable (Y_{ij}) that represents the directed relation from actor i to actor j [Robins et al., 2007].

$$Y_{ij} = \begin{cases} 1 & \text{if there is a directed tie from i to j} \\ 0 & \text{otherwise} \end{cases}$$

→ y_{ij} is the realized network tie of the random variable Y_{ij}

Let Y be the $N \times N$ matrix of all random variables (Y_{ij}) and let y be the realized (observed) network.

Exponential random graph model

The probability of observing the y network follows the exponential family distribution:

$$\Pr(Y = y) = k(\eta)^{-1} \exp\left[\sum_i \eta_i g_i(y)\right]$$

Exponential random graph model

The probability of observing the y network follows the exponential family distribution:

$$\Pr(Y = y) = k(\eta)^{-1} \exp\left[\underbrace{\sum_i \eta_i g_i(y)}\right]$$

- → $\sum_i$: sum over all network configurations.
- → $g_i(y)$: the counts of ith network configuration in the observed network.
- → η_i: parameters associated with the ith network configurations.

Configurations in exponential random graph models

Network configurations capture specific local structures [Snijders et al., 2006]. Examples:

- $S_1(y)$: the number of edges in y.
- $S_2(y)$: the number of two-stars in y.
- $D_k(y)$: the number of actors with degree k in y.
- $T(y)$: the number of triangles in y.
- $O(y)$: the number of isolates in y.

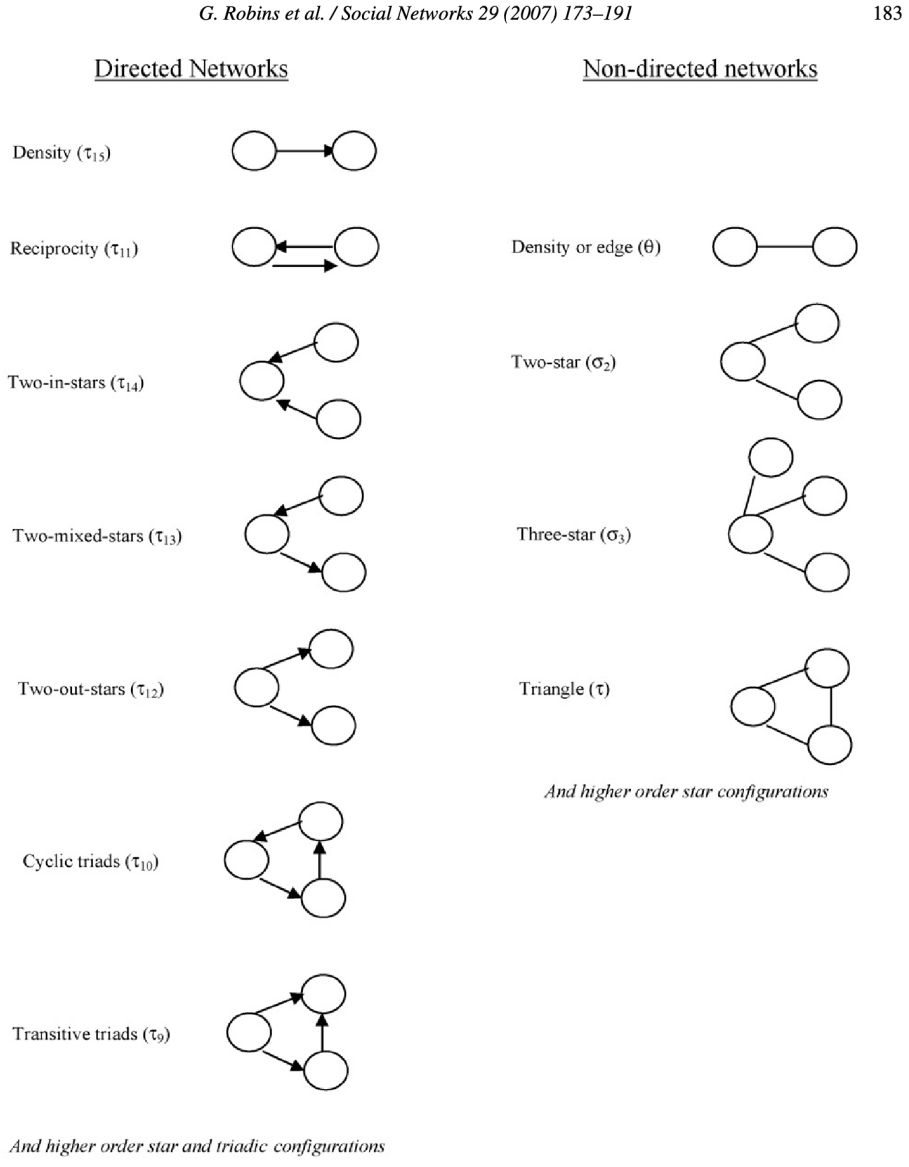


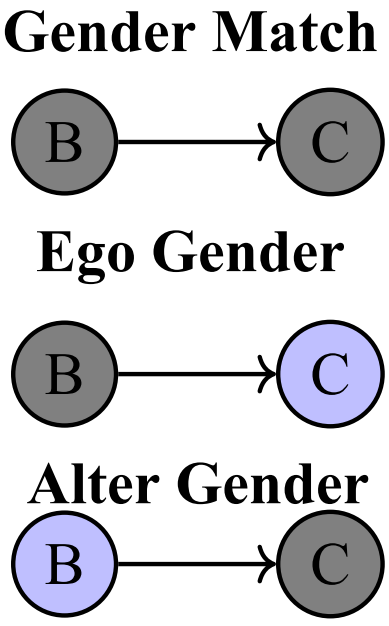
Fig. 1. Configurations and parameters for Markov random graph models.

Configurations in exponential random graph models

Networks configurations may also relate to exogenous actor-level attributes X.

Assume X is gender (male = 1; female = 0).

- → Match(y): the number of edges in y where the i and j actor have the same gender.
- → Ego$X(y)$: the number of edges in y where i is male.
- → Alter$X(y)$: the number of edges in y where j is male.



Configurations in exponential random graph models

Homogeneity assumption: We commonly assume homogeneity of the network configurations, which allows $g_i(y)$ to represent the count of the i configuration [Frank and Strauss, 1986, Robins et al., 2007]. If we assume heterogeneity, there would be a parameter value η for each occurrence of the i configuration.

"In considering reciprocity, Paul may tend very strongly to reciprocate friendship offers from others, but Mary might be more cautious. For the purpose of constructing a simpler model, however, we may assume that there is a single tendency for reciprocity shared by both Mary and Paul. The resulting error is then consumed into the model as statistical noise." [Robins et al., 2007, 180]

Exponential random graph model

$$\Pr(Y = y) = \underbrace{k(\eta)^{-1}} \exp\left[\sum_i \eta_i g_i(y)\right]$$

$k(\eta)$ is the normalizing constant that ensures the probability distribution sums to 1.

$$k(\eta) = \sum_{z \in y} \exp\left[\sum_i \eta_i g_i(z)\right]$$

The size of y is $2^{N(N-1)}$.

Exponential random graph models (tie-level)

The previous network-level probability formulation can be expressed at the individual tie-level [Strauss and Ikeda, 1990].

$$\text{logit}[\Pr(y_{ij} = 1 | C_{ij})] = \eta^T \delta_{ij}$$

where:

- $\text{logit}(t) = \ln(t/(1-t))$.
- C_{ij}: is the set of all realized ties beyond the (i, j) tie.
- δ_{ij}: represent change statistics.

Exponential random graph models (tie-level)

The change statistics δ_{ij} represent the change in the network configurations $g_i(y)$ when the (i, j) tie changes from 1 to 0.

$$\delta_{ij} = g_i(y_{ij}^{+}) - g_i(y_{ij}^{-})$$

where:

- y_{ij}^{+} is the realized network with the (i, j) tie set to 1.
- y_{ij}^{-} is the realized network with the (i, j) tie set to 0.

For example, the reciprocity change statistic is 1 if the y_{ji} tie is equal to 1 and 0 if not.

Exponential random graph models (tie-level)

Change statistic terms are either dyad-independent or dyad-dependent [Hunter et al., 2008b, Morris et al., 2008].

Dyad-independence:

"Definition 1: A dyadic independence term is a term in an ERGM for which the corresponding network change statistic may always be calculated, regardless of the values of i and j, without knowing anything about y except possibly (in the case of a directed network) the value of y_{ji}." [Hunter et al., 2008b, 11]
Examples: edges and exogenous actor-level covariates (gender homophily; see Morris et al. 2008).

Dyad-dependence:

Examples: indegree, outdegree, two-stars, transitive triads, three-cycles, etc. [Morris et al., 2008].

Maximum Likelihood Estimation

Following our above equations, the log-likelihood of an ERGM is:

$$\ell(\eta) = \eta^T g(y) - \log(k(\eta))$$

We wish to use Maximum Likelihood Estimation to find the set of η values that maximize the likelihood of the observed network y.

Main problem: the size of the normalizing constant, $k(\eta)$, is too big for modern computers for moderate networks.

- → if $N = 5$, $|k(\eta)|$ is 1,048,576.
- → if $N = 10$, $|k(\eta)|$ is greater than 1 trillion.

Maximum Pseudo Likelihood Estimation

Strauss and Ikeda [1990] and Wasserman and Pattison [1996] suggested to use Maximum Pseudo Likelihood Estimation (MPLE) to find the set of estimates. Instead of maximizing the joint probability distribution of the network, MPLE maximizes the product of conditional tie probabilities.

Recall that $\text{logit}[\Pr(y_{ij} = 1|C_{ij})] = \eta^T \delta_{ij}$. The MPL is the product of conditional tie probabilities:

$$\Pr(y|\eta) = \prod_{(i,j),\ i \neq j} \Pr(y_{ij} \mid C_{ij})$$

Maximum Pseudo Likelihood Estimation: Problems

The MPL ERGM fit via logistic regression routines assumes that the ties are conditionally independent given all other ties in the realized network. This assumption is often violated when modeling social networks.

As Hunter et al. [2008b, 254] points out, "a fundamental principle of social network analysis is that dependence among edges is a guiding force in the formation of networks; that is, the Y_{ij} that make up an adjacency matrix Y are not independent of one another."

Consequence of MPL ERGMs when the outcomes are not independent and/or a dyad-dependence term is included [van Duijn et al., 2009]:

- → The MPLE is not equal to the MLE.
- → The standard errors are too small making hypothesis tests and inferences invalid.

Monte Carlo Markov Chain Maximum Likelihood Estimation

Based on the early work of Frank and Strauss [1986], Handcock [2003] proposed to use Monte Carlo Markov Chain Maximum Likelihood Estimation (MCMC-MLE) to find the set of η estimates.

The basic idea is this: "if a large number of simulations from a social network in the same ballpark as that of the observed network can be generated then these can be used to approximate the normalization function to a desired accuracy." [Handcock, 2003, 9]

→ We can take samples from the normalizing constant to approximate the log-likelihood function!

Monte Carlo Markov Chain Maximum Likelihood Estimation

In a basic way, MCMC-MLE works as such:

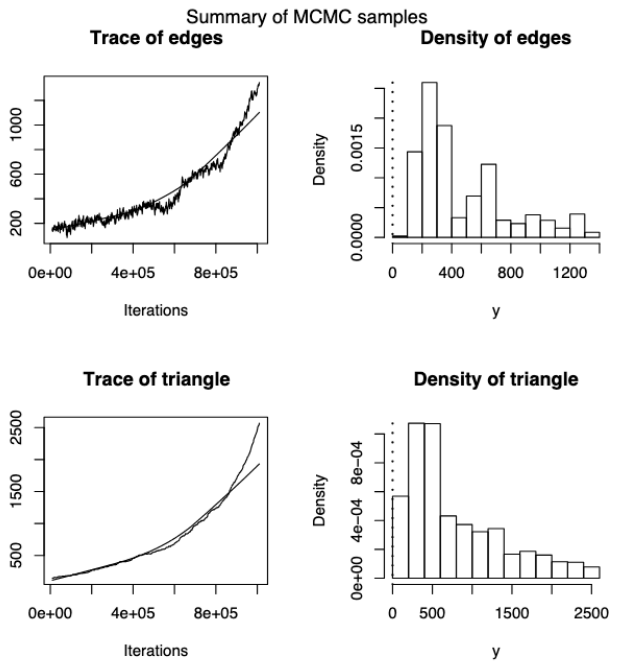
- → Based upon the current estimates of η^0, construct a Metropolis-Hasting Monte Carlo Markov Chain where we:
  - ☞ Randomly select an (i, j) tie from the current network y^0.
  - ☞ Compute the change statistic for the y^0_{ij} tie and change the tie with probability:
    $$\min\{1, \exp[\eta^0 \delta^0_{ij}]\}$$
  - ☞ Take a random sample of networks from the created Markov Chain of tie toggles.
  - ☞ Based upon the sample of networks, update the ML estimates η^0 using a Newton-Rhapson iteration.
- → Repeat the above procedure until adequate convergence of the log-likelihoods is achieved.

MCMC-MLE Degeneracy

Recall that MCMC-MLE is based on the idea that "if a large number of simulations from a social network in the same ballpark as that of the observed network can be generated then these can be used to approximate the normalization function to a desired accuracy." [Handcock, 2003, 9]

Degeneracy occurs "if the model places almost all its probability mass on a small number of graph configurations" and does not "simulate graphs close enough to produce realizations that cover the observed values of the statistics." [Handcock, 2003, 9]

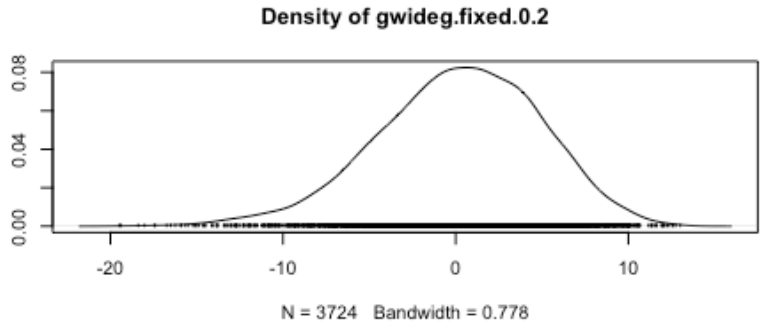
For example, the most common are (near) complete or empty graphs! (see Snijders et al. 2006 for a helpful discussion)



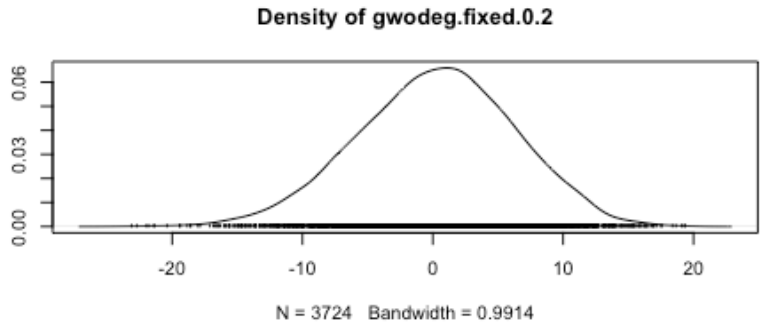
Source: Goodreau et al. [2008]

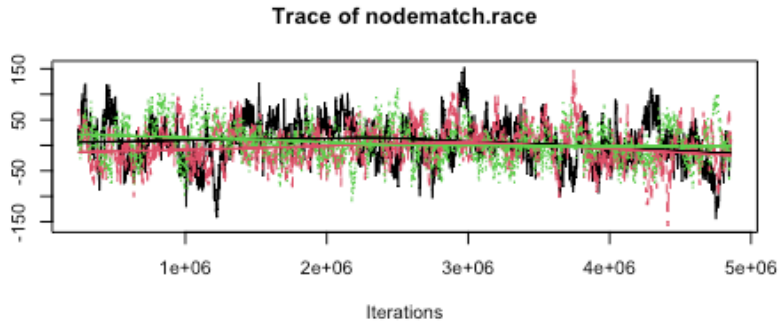
MCMC-MLE Degeneracy

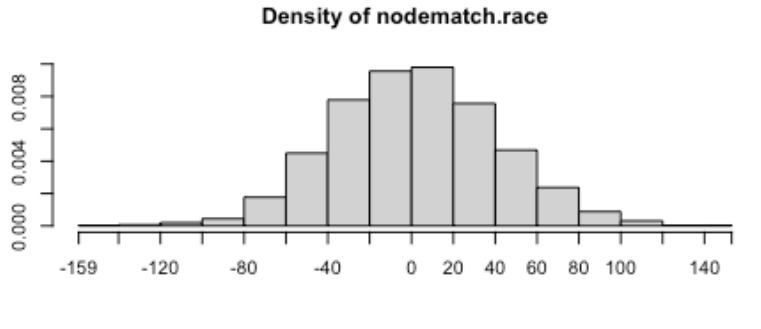












How to account for model degeneracy?

1. Increase the MCMC (a) sample size, (b) the intervals between consecutive samples, and/or (c) increase the burn-in value (the number of toggles before the first sample is taken).

2. Re-specify the current model by including or dropping terms that account for the divergence.

3. "Use a statistic that places decreasing weights on the higher degrees" [Snijders et al., 2006, 112]

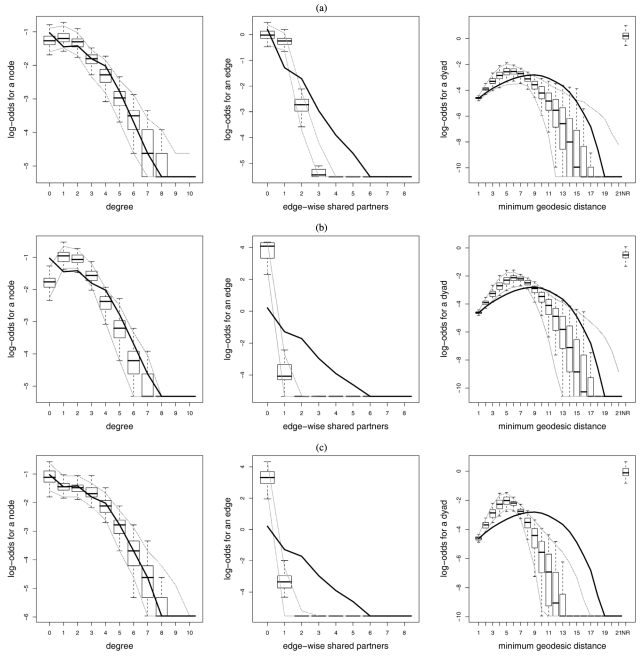
   → Geometrically weighted statistics, such as geometrically weighted edgewise shared partners [Snijders et al., 2006].

Post-estimation checklist

1. Assess the MCMC samples and check for model degeneracy.
2. Assess the ERGM goodness-of-fit [Hunter et al., 2008a]
   - → How well does the fitted model capture the underlying data generating process?
3. Check for the multicollinearity of effects by using the ERGM variance inflation factor [Duxbury, 2021]
   - → VIF values greater than 20 indicate moderate collinearity.
   - → VIF values greater than 100 indicate severe collinearity.

ERGM Goodness-of-Fit

"One way to develop an idea of how well a model fits is by **comparing a set of observed network statistics with the range of the same statistics obtained by simulating many networks from the fitted ERGM**. If the observed network is not typical of the simulated network for a particular statistic, then the model is either degenerate ... or poorly fitted." [Hunter et al., 2008a, 255]

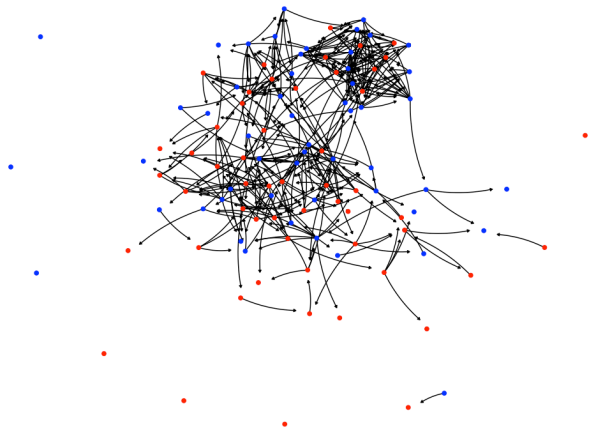


Source: Hunter et al. [2008a]

Faux Desert High School and the `statnet` R package

In this lecture, we will use the `statnet` R package to fit a set of ERGMs to the Faux Desert High School.

The Faux Desert High School is a simulated directed friendship network from a high school in the US with 107 adolescents based on the Add Health dataset.



Source: Hunter et al. [2008a]

Results of the Fitted Faux Desert High School ERGM

```
## Call:
## ergm(formula = faux.desert.high ~ edges + mutual + gwesp(decay = 0.2,
##     fixed = TRUE, type = "OTP") + gwidegree(decay = 0.2, fixed = TRUE) +
##     gwodegree(decay = 0.2, fixed = TRUE) + nodematch("race") +
##     nodematch("sex") + nodematch("grade") + nodeocov("sex") +
##     nodeicov("sex"), control = control.ergm(MCMC.samplesize = 10000,
##     MCMC.burnin = 10000, MCMC.interval = 1000, MCMLE.maxit = 30,
##     seed = 3718, parallel = 3, parallel.type = "PSOCK"))
##
## Monte Carlo Maximum Likelihood Results:
##
##                     Estimate Std. Error MCMC % z value Pr(>|z|)
## edges               -4.60926    0.18547      0 -24.852  < 1e-04 ***
## mutual               1.69420    0.19436      0   8.717  < 1e-04 ***
## gwesp.OTP.fixed.0.2  0.94639    0.10630      0   8.903  < 1e-04 ***
## gwideg.fixed.0.2     0.16415    0.36193      0   0.454 0.650156
## gwodeg.fixed.0.2    -1.14884    0.31807      0  -3.612 0.000304 ***
## nodematch.race       0.09513    0.08814      0   1.079 0.280478
## nodematch.sex        0.19217    0.08570      0   2.242 0.024942 *
## nodematch.grade      1.55037    0.11184      0  13.862  < 1e-04 ***
## nodeocov.sex        -0.19953    0.09123      0  -2.187 0.028726 *
## nodeicov.sex         0.03326    0.09966      0   0.334 0.738573
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 15723  on 11342  degrees of freedom
##  Residual Deviance:  2596  on 11332  degrees of freedom
##
## AIC: 2616  BIC: 2689  (Smaller is better. MC Std. Err. = 0.9479)
```

Results of the Fitted Faux Desert High School ERGM: Global Interpretation

```
Call:
ergm(formula = faux.desert.high ~ edges + mutual + gwesp(decay = 0.2,
    fixed = TRUE, type = "OTP") + gwidegree(decay = 0.2, fixed = TRUE) +
    gwodegree(decay = 0.2, fixed = TRUE) + nodematch("race") +
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---
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     Null Deviance: 15723  on 11342  degrees of freedom
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AIC: 2616  BIC: 2689  (Smaller is better. MC Std. Err. = 0.9479)
```

The observed network contains **more reciprocal ties and transitive closure** than expected by chance, net of all factors.

In addition, the observed network contains more **sex and grade homophilous ties** than expected by chance, net of all other factors.

Results of the Fitted Faux Desert High School ERGM: Tie Interpretation

```
Call:
ergm(formula = faux.desert.high ~ edges + mutual + gwesp(decay = 0.2,
    fixed = TRUE, type = "OTP") + gwidegree(decay = 0.2, fixed = TRUE) +
    gwodegree(decay = 0.2, fixed = TRUE) + nodematch("race") +
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---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

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AIC: 2616  BIC: 2689  (Smaller is better. MC Std. Err. = 0.9479)
```

The odds of the (i, j) tie being 1 is $\exp[1.694] = 5.442$ **times greater** if the (j, i) tie is 1, net of all other controls.

Results of the Fitted Faux Desert High School ERGM: Tie Interpretation

We can also use **average marginal effects (AMEs)** to interpret a fitted ERGM [Duxbury and Wertsching, 2023]. This is especially important since, like logistic regression estimates, ERGM estimates are scaled by an unknown constant.

```
ergm.AME(fitted.model,var1="mutual")
```

	AME	Delta SE	Z	P
mutual	0.043	0.005	8.717	0

The AME for reciprocity is 0.043 (SE: 0.005; p-value < 0.001), indicating that the probability of the (i, j) tie being 1 is increased by 0.043 if the (j, i) tie exists.

Results of the Fitted Faux Desert High School ERGM: Tie Interpretation

```
Call:
ergm(formula = faux.desert.high ~ edges + mutual + gwesp(decay = 0.2,
    fixed = TRUE, type = "OTP") + gwidegree(decay = 0.2, fixed = TRUE) +
    gwodegree(decay = 0.2, fixed = TRUE) + nodematch("race") +
    nodematch("sex") + nodematch("grade") + nodeocov("sex") +
    nodeicov("sex"), control = control.ergm(MCMC.samplesize = 10000,
    MCMC.burnin = 10000, MCMC.interval = 1000, MCMLE.maxit = 30,
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---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 15723  on 11342  degrees of freedom
 Residual Deviance:  2596  on 11332  degrees of freedom

AIC: 2616  BIC: 2689  (Smaller is better. MC Std. Err. = 0.9479)
```

The odds of the (i, j) tie being 1 is $\exp[0.94639] = 2.576$ **times greater** if the closes an additional triangle, however, this effect decreases as the number of triangles increase, net of all other controls.

The AME for geometrically weighted edgewise shared partners is 0.024 (SE: 0.003; p-value < 0.001), indicating that for each additional triangle closed, the probability of the (i, j) tie being 1 increases by 0.024.

Results of the Fitted Faux Desert High School ERGM: Tie Interpretation

```
Call:
ergm(formula = faux.desert.high ~ edges + mutual + gwesp(decay = 0.2,
    fixed = TRUE, type = "OTP") + gwidegree(decay = 0.2, fixed = TRUE) +
    gwodegree(decay = 0.2, fixed = TRUE) + nodematch("race") +
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    nodeicov("sex"), control = control.ergm(MCMC.samplesize = 10000,
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Monte Carlo Maximum Likelihood Results:

                   Estimate Std. Error MCMC % z value Pr(>|z|)
edges              -4.60926    0.18547      0 -24.852  < 1e-04 ***
mutual              1.69420    0.19436      0   8.717  < 1e-04 ***
gwesp.OTP.fixed.0.2 0.94639    0.10630      0   8.903  < 1e-04 ***
gwideg.fixed.0.2    0.16415    0.36193      0   0.454 0.650156
gwodeg.fixed.0.2   -1.14884    0.31807      0  -3.612 0.000304 ***
nodematch.race      0.09513    0.08814      0   1.079 0.280478
nodematch.sex       0.19217    0.08570      0   2.242 0.024942 *
nodematch.grade     1.55037    0.11184      0  13.862  < 1e-04 ***
nodeocov.sex       -0.19953    0.09123      0  -2.187 0.028726 *
nodeicov.sex        0.03326    0.09966      0   0.334 0.738573
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 15723  on 11342  degrees of freedom
 Residual Deviance:  2596  on 11332  degrees of freedom

AIC: 2616  BIC: 2689  (Smaller is better. MC Std. Err. = 0.9479)
```

The odds of the (i, j) tie being 1 is $\exp[1.55037] = 4.713$ **times greater** if i and j are in the same grade, net of all other controls.

The AME for grade homophily (nodematch) is 0.039 (SE: 0.003; p-value < 0.001), indicating that the probability of the (i, j) tie being 1 increases by 0.039 if the i and j are in the same grade, net of all other controls.

Possible extensions

Here we focused on applying ERGMs to directed one-mode networks. Possible extensions include:

- → undirected one-mode networks [Hunter et al., 2008b].
- → ego-centric networks [Salter-Townshend and Brendan Murphy, 2015].
- → valued ties [Huang and Butts, 2023].
- → bipartite (two-mode) networks [Wang et al., 2009].
- → multilevel networks [Wang et al., 2013, Breuer and Baradel, 2025]
- → longitudinal/panel networks [Hanneke et al., 2010, Krivitsky and Handcock, 2014, Leal, 2021].

Concluding remarks

The **exponential random graph model (ERGM)** is an inferential network model that allows us to understand how local social processes and structures lead to the emergence of global network structures.

We have looked over model interpretation and checking goodness-of-fit!

Next week, we will discuss **relational event models** (REMs)!



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